$E_{CFG} = \{\langle G\rangle \mid G \text{ is a CFG } \& \ L(G) = \phi\}$

Theorem 4.8] E_{CFG} is decidable.

ex. $S \to A$
$A \to S$
$T \to \#$

R = " On input $\langle G\rangle$
1. Mark all terminal symbols
2. Repeat until no new variables can be marked.
3. ↳ Mark any variable A where G has a rule $A \to U_1 U_2 \dots U_k$ and each U_i has been marked.
4. If the start variable is not marked, accept. Otherwise, reject. " □

$EQ_{CFG} = \{\langle G, H\rangle \mid G \ \& \ H \text{ are CFG's and } L(G) = L(H)\}$

Facts : * Regular languages are closed under star, union, intersection, complement.

* The intersection of a CFL and a regular language is context-free.

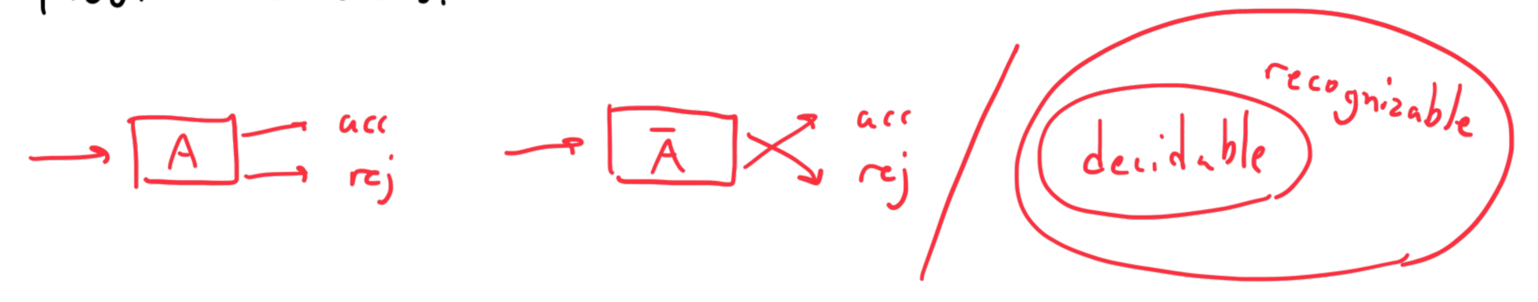
* CFL's are not closed under complement or intersection. They are closed under union, concatenation, and star.

EQ_{CFG} is not decidable. No similar technique such as for E_{DFA} can be employed.

Theorem 4.22] A language A is decidable iff it is recognizable and co-recognizable (that is, A & $\bar{A}$ are recognizable).

sufficiency

⇒ if language A is decidable then $A, \bar{A}$ are recognizable.
Proof: Obvious.



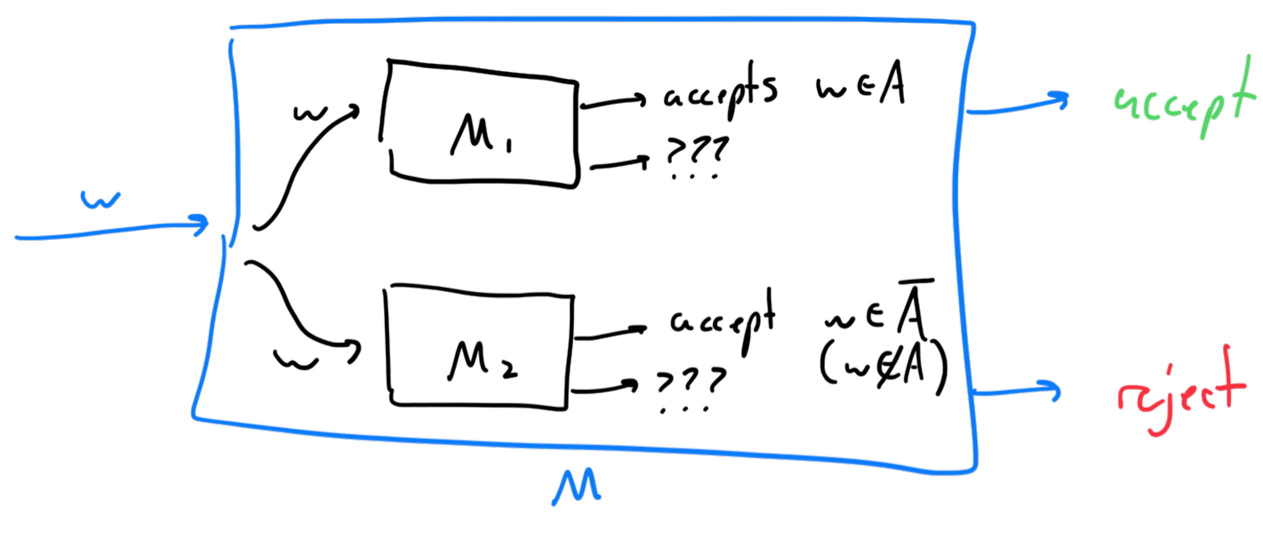
necessity

⇐ if $A, \bar{A}$ are recognizable then A is decidable.

Proof. Let M_1 & M_2 be recognizers for A & $\bar{A}$.

M = " On input w :
Loop i : 1, 2, 3 ....
If M_1 halts on w in 'i' steps in the accept state, accept.
If M_2 halts on w in 'i' steps in the accept state, reject.

Thus, we are running M_1 & M_2 in parallel. Two tapes exist, M takes turns simulating M_1 & M_2. Either M_1 or M_2 accepts w. M accepts all strings in A and rejects strings not in A. Therefore, we are done. □



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why not finish M_1 before starting M_2?
M_1 may never halt! It is only a recognizer.